

Higher-Dimensional Completion Geometry for Linear Counting Languages

Shortest-Prefix Quotients, Rank-One Topology, and Eventual Geodesic Structure

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Abstract

Let $A \in \mathbb{Z}^{k \times d}$ and let

$$L_A = \{w \in \Sigma^* : A\Psi(w) = 0\},$$

where Ψ is the Parikh map. Under the completion-totality assumption that $Ah = 0$ for some strictly positive $h \in \mathbb{N}^d$, every prefix residual has a finite minimum completion. We study the geometry of all shortest completions rather than recognizability alone.

For a residual r , let $\ell_A(r)$ be its minimum completion length and let $O_A(r)$ be the set of optimal Parikh vectors. We prove that the finite down-set $\mathcal{C}_A(r) = \downarrow O_A(r)$ is exactly the set of Parikh vectors of prefixes of shortest completions:

$$u \in \mathcal{C}_A(r) \iff \ell_A(r + Au) = \ell_A(r) - |u|_1.$$

Writing $K_A = \ker_{\mathbb{Z}}[A; \mathbf{1}^\top]$, the complete filled shortest-path interval from r to 0 is the cubical quotient

$$\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A.$$

It is a literal coordinate box exactly when the optimal Parikh vector is unique. If $\text{rank}_{\mathbb{Z}} K_A = 1$, the quotient has a sharp global homotopy dichotomy:

$$\mathfrak{I}_A(r) \simeq * \quad \text{for a unique optimum,} \quad \mathfrak{I}_A(r) \simeq S^1 \quad \text{otherwise.}$$

The proof constructs a periodic cubical saturation, proves it contractible by a good-cover argument, and identifies the quotient as a $K(\mathbb{Z}, 1)$.

The local compatibility complex of geodesic letter actions is exactly the squarefree divisor complex of the augmented optimal semigroup element. In rank at most one every connected component of this local complex is contractible, while a four-letter rank-two example already has nontrivial first homology. Thus scalar systems exhibit a sharp $d = 2, 3, 4+$ hierarchy: literal geometry, the first global cycle, and the first possible local holes. Finally, along every residual ray we obtain an eventual affine defect profile; its finite circular symmetry determines the exact minimal eventual distance period and the eventual periods of local geodesic observables. The contribution is the coupling of minimum-completion fibers, shortest-prefix schedules, residual quotients, and equal-length lattice identifications; standard affine-semigroup, squarefree-divisor, Graver/Markov, state-complex, Squier-complex, and vector-partition constructions are used only as neighboring background.

Keywords: deterministic pushdown automata; counting languages; shortest completions; affine semigroups; cubical complexes; squarefree divisor complexes; geodesics; factorization fibers.

1 Introduction

The binary counting languages

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}, \quad \gcd(p,q) = 1,$$

have an elementary residual graph: reading 0 subtracts p and reading 1 adds q . In that two-letter setting a residual has a unique minimum Parikh demand, and the orderings of that demand form a coordinate rectangle of shortest paths. Earlier companion manuscripts developed this completion viewpoint directly from a concrete deterministic pushdown design and used it to derive exact binary rectangle and representation results [13, 14].

A first structural obstruction relative to the binary case is that a minimum completion need not have a unique Parikh vector. A residual may admit several optimal integer factorizations of equal length, and their coordinate boxes overlap and collide after projection to the residual graph. This raises three questions.

- (i) Which partial letter schedules occur on shortest completions?
- (ii) How should multiple optimal Parikh vectors be identified to recover the actual shortest residual interval?
- (iii) What topology can be created by those identifications, and how does it depend on the rank of the equal-length trade lattice?

We answer these questions for integer matrix counting constraints. The central object is not the factorization fiber alone, but the sequence

$$\text{minimum-completion fiber} \longrightarrow \text{shortest-prefix down-set} \longrightarrow \text{equal-length lattice quotient}.$$

The quotient retains all coordinatewise prefixes of all minimum completions and identifies exactly those prefixes that represent the same residual at the same geodesic depth.

Main results. The paper establishes the following package.

1. The geodesic-prefix criterion gives an exact intrinsic test for membership in $\downarrow O_A(r)$.
2. The filled shortest residual interval is the cubical quotient $|\downarrow O_A(r)|/K_A$, where $K_A = \ker_{\mathbb{Z}}[A; \mathbf{1}^\top]$.
3. The quotient is a literal coordinate box if and only if the optimal Parikh vector is unique.
4. In rank one, the global quotient is contractible in the unique case and homotopy equivalent to S^1 in every nonunique case.
5. The local action-compatibility complex is the classical squarefree divisor complex of the augmented optimal element. Its components are contractible in rank at most one, whereas rank two already permits a genuine local H_1 hole.
6. In the scalar case, rank $K = d-2$, yielding a sharp $d = 2, 3, 4+$ hierarchy of shortest-geodesic phenomena.
7. Along a lattice ray, a finite stabilized reduced-cost profile controls eventual completion distance and local geodesic geometry; the profile gives the exact minimal eventual affine period.

Novelty boundary. Several broad ingredients have substantial prior literatures. Equal-length factorizations and factorization invariants in half-factorial affine semigroups are established [1]; Graver and Markov moves are standard tools for integer fibers [2, 3]; squarefree divisor complexes are standard semigroup complexes [4]; factorization support/trade graphs and factorization-derived cubical complexes are also established [5]; independent moves are filled cubically in state-complex and Squier-complex theories [8, 9, 10]; and vector partition functions have classical chamberwise quasipolynomial structure [11]. We do not claim novelty for those general constructions. The candidate contribution is their specific meeting point in minimum-completion geodesics: the shortest-prefix down-set, its equal-length residual quotient, and the rank-one global topology. We are not aware of a previous theorem identifying this exact shortest-prefix quotient or establishing the rank-one homotopy dichotomy.

2 Matrix counting languages and completion totality

Let $\Sigma = \{a_1, \dots, a_d\}$ and let

$$\Psi : \Sigma^* \rightarrow \mathbb{N}^d$$

be the Parikh map. Fix $A \in \mathbb{Z}^{k \times d}$ and define

$$\rho(w) = A\Psi(w), \quad L_A = \{w : \rho(w) = 0\}.$$

Residuals lie in the lattice $A\mathbb{Z}^d$.

Definition 2.1 (Completion totality). *We say that A is completion-total if there exists $h \in \mathbb{N}_{>0}^d$ with*

$$Ah = 0.$$

This convenient hypothesis makes every lattice residual completable.

Lemma 2.2 (Every residual has a nonnegative completion). *Assume completion totality. For every $r \in A\mathbb{Z}^d$ there exists $x \in \mathbb{N}^d$ with $Ax = -r$.*

Proof. Write $r = Az$ with $z \in \mathbb{Z}^d$. For sufficiently large t , $th - z \in \mathbb{N}^d$. Since $Ah = 0$,

$$A(th - z) = -Az = -r.$$

□

Proposition 2.3 (Residuals classify prefix quotients). *Under completion totality, for prefixes $u, v \in \Sigma^*$,*

$$u^{-1}L_A = v^{-1}L_A \iff \rho(u) = \rho(v).$$

Proof. A continuation z completes u exactly when $A\Psi(z) = -\rho(u)$, so equal residuals give equal quotients. Conversely, if $\rho(u) \neq \rho(v)$, choose by theorem 2.2 a nonnegative Parikh vector x with $Ax = -\rho(u)$ and realize it by some word z . Then $uz \in L_A$ but $vz \notin L_A$. □

For $r \in A\mathbb{Z}^d$, define the minimum completion length

$$\ell_A(r) = \min\{\mathbf{1}^\top x : x \in \mathbb{N}^d, Ax = -r\}$$

and the optimal Parikh fiber

$$O_A(r) = \{x \in \mathbb{N}^d : Ax = -r, \mathbf{1}^\top x = \ell_A(r)\}.$$

Introduce the augmented matrix

$$C = \begin{bmatrix} A \\ \mathbf{1}^\top \end{bmatrix}, \quad K_A = \ker_{\mathbb{Z}} C.$$

The augmented generators $C_i = (A_i, 1)^\top$ generate a half-factorial affine semigroup: every factorization of one augmented element has the same length, namely its last coordinate. In particular $O_A(r)$ is exactly the factorization fiber of

$$b_r = (-r, \ell_A(r))$$

in this augmented semigroup. We use this observation for comparison with factorization theory, not as a new algebraic construction.

2.1 Continuous optimum geometry

Define

$$P_r^A = \{x \in \mathbb{R}_{\geq 0}^d : Ax = -r, \mathbf{1}^\top x = \ell_A(r)\}.$$

Let

$$S(r) = \{i : \exists x \in P_r^A \text{ with } x_i > 0\}$$

be its active support.

Proposition 2.4 (Dimension of the continuous optimum polytope). *If $P_r^A \neq \emptyset$, then*

$$\dim P_r^A = |S(r)| - \text{rank}_{\mathbb{R}} \begin{bmatrix} A_{S(r)} \\ \mathbf{1}^\top \end{bmatrix}.$$

Proof. For each $i \in S(r)$ choose a feasible point positive in coordinate i . Their average lies in the relative interior of the nonnegative orthant restricted to $S(r)$ and is positive on every active coordinate. Thus no further coordinate inequality is active on the relative interior. The dimension is therefore the number of active variables minus the rank of the affine equality system restricted to those variables. $\square$

This separates continuous degeneracy from integer geodesic multiplicity: P_r^A may have positive real dimension while $O_A(r)$ is a singleton, or conversely an integer fiber can expose discrete trade structure not visible from dimension alone.

3 Minimum-completion fibers and shortest prefixes

For $x \in \mathbb{N}^d$ write

$$B_x = \{u \in \mathbb{N}^d : 0 \leq u \leq x\}$$

coordinatewise. Define the finite down-set

$$\mathcal{C}_A(r) = \downarrow O_A(r) = \bigcup_{x \in O_A(r)} B_x.$$

Theorem 3.1 (Exact geodesic-prefix criterion). *For every $u \in \mathbb{N}^d$,*

$$u \in \mathcal{C}_A(r) \iff \ell_A(r + Au) = \ell_A(r) - |u|_1.$$

Equivalently, $\mathcal{C}_A(r)$ is exactly the set of Parikh vectors that occur as prefixes of shortest completions of residual r .

Proof. If $u \leq x$ for some $x \in O_A(r)$, then $x - u$ completes $r + Au$ in length $\ell_A(r) - |u|_1$. A shorter completion would concatenate with u to give a completion of r shorter than $\ell_A(r)$.

Conversely, suppose the displayed equality holds and let y optimally complete $r + Au$. Then

$$A(u + y) = -r, \quad |u + y|_1 = |u|_1 + \ell_A(r + Au) = \ell_A(r),$$

so $u + y \in O_A(r)$ and $u \leq u + y$. Thus $u \in \mathcal{C}_A(r)$. $\square$

Two immediate operational tests are useful.

Corollary 3.2 (First actions and compatible action sets). *For $i \in [d]$,*

$$i \text{ can be the first letter of a shortest completion } \iff \ell_A(r + Ae_i) = \ell_A(r) - 1.$$

More generally, for $J \subseteq [d]$,

$$\exists x \in O_A(r) \text{ with } x_j > 0 \ \forall j \in J \iff \ell_A(r + A\mathbf{1}_J) = \ell_A(r) - |J|.$$

Proof. Apply theorem 3.1 to e_i and to the squarefree vector $\mathbf{1}_J$. □

4 The filled shortest-geodesic quotient

We now retain not only the vertices of the down-set but also its commuting coordinate cells.

Definition 4.1 (Cubical realization of a down-set). *For a finite coordinatewise down-set $D \subseteq \mathbb{N}^d$, let $|D|$ be the cubical complex whose vertices are the elements of D and whose elementary J -oriented cube based at u is present exactly when*

$$u + \mathbf{1}_J \in D.$$

Every such cube records a set of distinct letter actions that may be executed in any order while remaining geodesic.

Define an equivalence on cells of $|\mathcal{C}_A(r)|$ by equal-length kernel translation: two equally oriented cells are equivalent when their base vertices differ by an element of K_A . Let

$$\mathfrak{I}_A(r) = |\mathcal{C}_A(r)| / K_A$$

denote the resulting finite cubical CW quotient.

Theorem 4.2 (Shortest-prefix quotient theorem). *The 0-skeleton of $\mathfrak{I}_A(r)$ is label-preserving isomorphic to the complete directed shortest-path interval from residual r to 0. Under this identification, elementary cubes are exactly compatible commuting sets of geodesic one-letter actions.*

Proof. Map a prefix vector to its reached residual,

$$\pi_r : \mathcal{C}_A(r) \rightarrow A\mathbb{Z}^d, \quad \pi_r(u) = r + Au.$$

By theorem 3.1, every image vertex lies on a shortest path to 0. Conversely every geodesic prefix has a Parikh vector in $\mathcal{C}_A(r)$.

If $\pi_r(u) = \pi_r(v)$, then $A(u - v) = 0$. Since both u and v are geodesic prefixes reaching the same residual,

$$\ell_A(r) - |u|_1 = \text{ell}_A(\pi_r(u)) = \text{ell}_A(\pi_r(v)) = \ell_A(r) - |v|_1,$$

so $\mathbf{1}^\top(u - v) = 0$ and therefore $u - v \in K_A$. The converse is immediate. Hence the vertex fibers are precisely the K_A -classes.

A J -cube at u exists exactly when $u + \mathbf{1}_J \in \mathcal{C}_A(r)$. By theorem 3.1, every subword using the distinct actions in J remains on a shortest path, and addition of coordinate vectors commutes. Kernel translation preserves orientation and residual labels, so the cell quotient gives precisely the filled commuting geodesic structure. □

4.1 When is the quotient literally a box?

Theorem 4.3 (Literal-box criterion).

$$|O_A(r)| = 1 \iff \mathfrak{I}_A(r) \text{ is a literal coordinate box.}$$

Proof. Suppose first that $O_A(r) = \{x\}$. Then $\mathcal{C}_A(r) = B_x$. If distinct $u, v \leq x$ were identified, then $u - v \in K_A$. The vector

$$x - u + v$$

is nonnegative, has the same residual contribution and the same length as x , and differs from x ; this would be a second element of $O_A(r)$. Thus no nontrivial kernel collision occurs in B_x , and the quotient is the literal box $|B_x|$.

Conversely, if $x \neq y$ belong to $O_A(r)$, then $x, y \in \mathcal{C}_A(r)$, both reach residual 0, and $x - y \in K_A$. Hence the quotient identifies distinct maximal geodesic schedules. It is therefore not a literal coordinate box with distinct coordinate vertices. $\square$

The theorem isolates the first source of global nontriviality: a kernel collision among shortest prefixes is not merely a local accident; it lifts to nonuniqueness of the optimal Parikh fiber.

5 Local factorization topology

The local compatibility data have a standard semigroup interpretation. Define

$$\mathcal{H}_A(r) = \{J \subseteq [d] : \exists x \in O_A(r), x_j > 0 \forall j \in J\}.$$

By theorem 3.2, its faces are exactly the sets of distinct letter types that can coexist on one shortest completion.

Let $S_C = \langle C_1, \dots, C_d \rangle$ be the affine semigroup generated by the augmented columns $C_i = (A_i, 1)^\top$, and recall $b_r = (-r, \ell_A(r))$.

Proposition 5.1 (Squarefree divisor identification).

$$\mathcal{H}_A(r) = \{J \subseteq [d] : b_r - \sum_{j \in J} C_j \in S_C\}.$$

Thus $\mathcal{H}_A(r)$ is exactly the squarefree divisor complex of b_r with respect to the augmented generators.

Proof. The semigroup membership on the right means that there is $y \in \mathbb{N}^d$ with

$$Cy = b_r - \sum_{j \in J} C_j.$$

Equivalently $x = y + \mathbf{1}_J$ satisfies $Cx = b_r$, i.e. $x \in O_A(r)$ and $x_j \geq 1$ for every $j \in J$. $\square$

Squarefree divisor complexes are established objects in the homological study of semigroup rings [4]; the new role here is their exact interpretation as local shortest-action compatibility.

5.1 Rank at most one: local topology is trivial

Theorem 5.2 (Rank-one local action topology). *If $\text{rank}_{\mathbb{Z}} K_A \leq 1$, then every connected component of $\mathcal{H}_A(r)$ is contractible. In particular,*

$$\tilde{H}_j(\mathcal{H}_A(r); \mathbb{Z}) = 0 \quad (j \geq 1).$$

Proof. If $\text{rank } K_A = 0$, the optimal fiber is a singleton $\{x\}$, so $\mathcal{H}_A(r)$ is the simplex on $\text{supp}(x)$.

Assume $K_A = \mathbb{Z}g$. The complete optimal fiber is an arithmetic interval

$$O_A(r) = \{x_t = x_0 + tg : 0 \leq t \leq N\}.$$

For each action i , the set

$$T_i = \{t \in [0, N] : x_t(i) > 0\}$$

is an integer interval because the coordinate $x_t(i)$ is affine in t . A set J is a face of $\mathcal{H}_A(r)$ exactly when $\bigcap_{j \in J} T_j \neq \emptyset$. Thus $\mathcal{H}_A(r)$ is the nerve of a finite family of intervals. By the nerve theorem, each connected component is homotopy equivalent to a connected component of their union, hence to an interval. $\square$

5.2 Rank two already permits a local hole

Example 5.3 (A four-letter H_1 obstruction). *Consider the scalar weight vector*

$$\omega = (-11, -2, 1, 10), \quad r = 7.$$

The optimal Parikh vectors are

$$(0, 4, 1, 0), \quad (1, 0, 4, 0), \quad (1, 3, 0, 1).$$

The maximal faces of $\mathcal{H}(r)$ are

$$\{2, 3\}, \quad \{1, 3\}, \quad \{1, 2, 4\}.$$

Hence

$$f_0 = 4, \quad f_1 = 5, \quad f_2 = 1.$$

The complex is connected and its unique 2-simplex does not fill the remaining graph cycle, so

$$\beta_1 = 1, \quad H_1(\mathcal{H}(r); \mathbb{Z}) \cong \mathbb{Z}.$$

The augmented equal-length kernel has rank two.

This example will become the sharp scalar threshold in section 7.

6 Rank-one global topology

The local complex can be topologically trivial even when the full shortest-prefix quotient is not. We now determine the global rank-one homotopy type exactly.

Assume

$$K_A = \mathbb{Z}g$$

with primitive nonzero generator g . Write

$$g = g^+ - g^-, \quad g^+, g^- \in \mathbb{N}^d, \quad \text{supp}(g^+) \cap \text{supp}(g^-) = \emptyset.$$

The complete optimal fiber is

$$O_A(r) = \{x_t = x_0 + tg : 0 \leq t \leq N\}.$$

Let

$$X = |\mathcal{C}_A(r)| = \left| \bigcup_{t=0}^N B_{x_t} \right|.$$

As a cubical down-set, X is contractible.

6.1 Overlap complexes

For $m \geq 1$, define

$$P_m = \{\sigma \subseteq X : \sigma + mg \subseteq X\},$$

where σ ranges over cubical cells and translation preserves cell orientation.

Lemma 6.1 (Exact m -step overlap). *For $1 \leq m \leq N$, translation by $-mg^-$ identifies P_m with the anchored cubical down-set*

$$\bigcup_{t=0}^{N-m} B_{x_t - mg^-}.$$

Hence P_m is nonempty and contractible. For $m > N$, $P_m = \emptyset$.

Proof. Suppose a cell σ and its translate $\sigma + mg$ both lie in X . Choose s with $\sigma + mg \subseteq B_{x_s}$. Then

$$\sigma \leq x_s - mg = x_{s-m}.$$

Completeness of the fiber forces $s \geq m$. Moreover $\sigma + mg \geq 0$ implies that the coordinates of σ on $\text{supp}(g^-)$ are at least those of mg^- . Subtracting mg^- therefore gives a cell in one of the anchored boxes $B_{x_{s-m} - mg^-}$. The converse follows by reversing the translation.

If $m > N$ and such a pair existed, two geodesic prefixes separated by mg would lift, by the collision argument behind theorem 4.3, to two optimal Parikh vectors separated by mg , contradicting the complete parameter span $0 \leq t \leq N$. $\square$

6.2 Periodic saturation and a good cover

Define

$$\tilde{X} = \bigcup_{k \in \mathbb{Z}} (X + kg) \subseteq \mathbb{R}^d, \quad X_k = X + kg.$$

Translation by g acts freely and cellularly on $\tilde{X}$, and

$$\tilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r).$$

Indeed, two cells of the fundamental region X become equal in the orbit quotient exactly when their bases differ by an integer multiple of g , which is precisely the rank-one equal-length kernel identification.

Lemma 6.2 (Good-cover intersections). *Let $k_0 < \dots < k_s$. Then*

$$X_{k_0} \cap \dots \cap X_{k_s} \neq \emptyset \iff k_s - k_0 \leq N.$$

Every nonempty finite intersection is contractible.

Proof. For a fixed cell, the set of translation indices k for which it belongs to X_k is an integer interval: the coordinate inequalities defining membership in the translated rank-one down-set each impose an interval constraint. Therefore membership in the two extreme translates implies membership in every intermediate translate. After translating by $-k_0 g$, the intersection of the extremes is the $(k_s - k_0)$ -step overlap complex from theorem 6.1, which is contractible exactly when $k_s - k_0 \leq N$ and empty otherwise. $\square$

The nerve of the cover $\{X_k\}_{k \in \mathbb{Z}}$ has vertex set $\mathbb{Z}$, and a finite set spans a simplex exactly when its diameter is at most N . For $N \geq 1$ this is the nerve of the interval cover

$$\{[k, k + N]\}_{k \in \mathbb{Z}}$$

of $\mathbb{R}$. The union is $\mathbb{R}$ and every finite nonempty intersection is an interval. Hence the nerve is contractible.

Theorem 6.3 (Contractible rank-one saturation). *If $N \geq 1$, then*

$$\tilde{X} \simeq *.$$

Proof. By theorem 6.2, $\{X_k\}$ is a locally finite cover by cubical subcomplexes with all nonempty finite intersections contractible. We use the standard CW/subcomplex form of the nerve lemma (equivalently, one may replace the locally finite subcomplex cover by compatible open regular neighborhoods without changing the intersection nerve) [6]. Its nerve is contractible by the interval-cover description above, hence $\tilde{X} \simeq *$. $\square$

Theorem 6.4 (Rank-one global homotopy dichotomy). *If $\text{rank}_{\mathbb{Z}} K_A = 1$, then*

$$\mathfrak{I}_A(r) \simeq \begin{cases} *, & |O_A(r)| = 1, \\ S^1, & |O_A(r)| > 1. \end{cases}$$

Proof. If the optimum is unique, theorem 4.3 gives a literal box, hence a contractible complex.

Assume $N \geq 1$. The action of $\mathbb{Z}$ by translation on $\tilde{X}$ is free, properly discontinuous, and cellular: a compact set meets only finitely many of its integer translates because $g \neq 0$. By theorem 6.3, $\tilde{X}$ is contractible, so

$$\tilde{X} \longrightarrow \tilde{X}/\mathbb{Z} \cong \mathfrak{I}_A(r)$$

is a universal cover with deck group $\mathbb{Z}$ by the standard covering-space action criterion [7]. Hence $\pi_1(\mathfrak{I}_A(r)) \cong \mathbb{Z}$ and all higher homotopy groups vanish. Thus $\mathfrak{I}_A(r)$ is a $K(\mathbb{Z}, 1)$, and therefore is homotopy equivalent to S^1 . $\square$

Combining theorems 5.2 and 6.4 yields a strict local/global separation: in the nonunique rank-one case every connected component of the squarefree-divisor action complex is contractible, while the full schedule-identification quotient carries a global circle.

7 Scalar hierarchy and exact three-letter formulas

Consider a scalar weight vector

$$\omega = (\omega_1, \dots, \omega_d) \in \mathbb{Z}^d$$

with at least two distinct entries, and set $A = \omega$ as a $1 \times d$ matrix. Then the two rows of

$$C = \begin{bmatrix} \omega \\ \mathbf{1}^\top \end{bmatrix}$$

are linearly independent, so

$$\text{rank}_{\mathbb{Z}} K_\omega = d - 2.$$

Consequently:

d	shortest-geodesic structure
2	rank 0: every optimum is unique and every shortest interval is a literal rectangle/box;
3	rank 1: the optimum fiber is an arithmetic interval and the global quotient is a box or S^1 , while every local action-complex component is contractible;
≥ 4	higher-rank trade geometry becomes possible; already at $d = 4$ theorem 5.3 gives a local complex with nontrivial H_1 .

Thus $d = 3$ is the first scalar alphabet size that can create a global cycle, and $d = 4$ is the first at which a local shortest-action hole can occur.

7.1 Three-letter rank-one normal form

Assume

$$\omega_1 < \omega_2 < \omega_3.$$

Put

$$\begin{aligned} a &= \omega_3 - \omega_2, & b &= \omega_2 - \omega_1, & h &= \gcd(a, b), \\ \alpha &= a/h, & \beta &= b/h, & m &= \alpha + \beta. \end{aligned}$$

A primitive equal-length kernel generator is

$$g = (\alpha, -m, \beta).$$

Write the complete optimal fiber as

$$O_A(r) = \{x(t) = y + tg : 0 \leq t \leq N\}, \quad y = (y_1, y_2, y_3).$$

Let

$$z = y_2 - mN, \quad 0 \leq z < m,$$

and

$$P_t = (y_1 + \alpha t + 1)(y_3 + \beta t + 1).$$

Slicing the prequotient down-set by the middle coordinate gives

$$|\mathcal{C}_A(r)| = m \sum_{t=0}^{N-1} P_t + (z+1)P_N.$$

Contiguity of every rank-one orbit in the down-set then yields the quotient vertex count.

Proposition 7.1 (Exact three-letter quotient vertex count). *Let $\Omega(y, N)$ denote the number of vertices of $\mathcal{C}_A(r)/\mathbb{Z}g$. If $N = 0$, then*

$$\Omega(y, 0) = (y_1 + 1)(y_2 + 1)(y_3 + 1).$$

If $N \geq 1$, then

$$\boxed{\Omega(y, N) = mP_{N-1} + (z+1)(P_N - P_{N-1}).}$$

Proof. For each orbit, the intersection with $\mathcal{C}_A(r)$ is an integer interval. An orbit containing s vertices contributes $s - 1$ adjacent pairs $u, u + g$. Therefore the number of orbits equals the number of vertices minus the number of adjacent pairs. The same middle-coordinate slicing gives

$$\#\{u \in \mathcal{C}_A(r) : u + g \in \mathcal{C}_A(r)\} = m \sum_{t=0}^{N-2} P_t + (z+1)P_{N-1},$$

and subtraction gives the formula. □

The same reduction applies orientation-by-orientation to elementary cubes. For $J \subseteq \{1, 2, 3\}$ let

$$T_J = \{t \in [0, N] \cap \mathbb{Z} : x_i(t) \geq 1 \ \forall i \in J\}.$$

This is an integer interval $[L_J, U_J]$, possibly empty. If nonempty put

$$N_J = U_J - L_J, \quad y^{(J)} = x(L_J) - \mathbf{1}_J.$$

Proposition 7.2 (Exact rank-one cubical f -vector). *The number of J -oriented quotient cubes is*

$$f_J(r) = \begin{cases} 0, & T_J = \emptyset, \\ \Omega(y^{(J)}, N_J), & T_J \neq \emptyset. \end{cases}$$

Hence

$$f_j(r) = \sum_{|J|=j} f_J(r), \quad 0 \leq j \leq 3.$$

Proof. A J -cube based at u is present exactly when $u + \mathbf{1}_J \in \mathcal{C}_A(r)$. Restricting the optimal fiber to parameters supporting all coordinates in J and subtracting $\mathbf{1}_J$ produces another rank-one down-set with endpoint data $(y^{(J)}, N_J)$. Translation by g identifies two equally oriented cubes exactly when it identifies their base vertices, so theorem 7.1 applies. $\square$

Example 7.3. *For*

$$\omega = (-5, -3, 1), \quad r = 9,$$

one has $\alpha = 2$, $\beta = 1$, $m = 3$ and

$$O_A(9) = \{(0, 3, 0), (2, 0, 1)\}.$$

The quotient cubical f -vector is

$$(f_0, f_1, f_2, f_3) = (8, 10, 2, 0).$$

Thus this shortest residual interval has eight vertices, ten labeled edges, and two elementary commuting squares, while theorem 6.4 says its filled quotient is homotopy equivalent to S^1 .

8 Eventual arithmetic geometry

The finite quotient geometry above varies with the residual. Along a fixed lattice ray, however, the minimum-completion problem develops an exact finite defect profile.

Fix $r, b \in A\mathbb{Z}^d$ and

$$r_n = r + nb, \quad n \in \mathbb{N}.$$

Consider the continuous directional program

$$\lambda_b = \min\{\mathbf{1}^\top x : Ax = -b, x \in \mathbb{R}_{\geq 0}^d\}.$$

Completion totality makes this feasible, and rationality gives $\lambda_b \in \mathbb{Q}_{\geq 0}$.

Choose a rational dual optimum y^* for

$$\max\{-b^\top y : A^\top y \leq \mathbf{1}\}.$$

Let L clear denominators so that the reduced costs

$$\delta_i = L(1 - A_i^\top y^*)$$

are nonnegative integers. For an optimal integer completion of r_n , define the scaled integrality defect

$$\Delta_r(n) = L(\ell_A(r_n) + (y^*)^\top r - n\lambda_b) \in \mathbb{N}.$$

8.1 Zero-defect shifts and the canonical period

Define

$$\mathcal{S}_b = \{t \geq 1 : \exists z \in \mathbb{N}^d, Az = -tb, |z|_1 = t\lambda_b\}.$$

A cleared-denominator rational primal optimum shows that $\mathcal{S}_b$ is nonempty, and it is closed under addition. Put

$$\tau_b = \gcd \mathcal{S}_b.$$

For $t \in \mathcal{S}_b$, the corresponding z has zero reduced cost and may be appended to a completion without changing its defect.

Theorem 8.1 (Stabilized defect profile). *For each residue class $s \in \mathbb{Z}/\tau_b\mathbb{Z}$, define*

$$M_s = \min\{\Delta_r(n) : n \geq 0, n \equiv s \pmod{\tau_b}\}.$$

Then for every s there is N_s such that

$$\Delta_r(n) = M_s$$

whenever $n \geq N_s$ and $n \equiv s \pmod{\tau_b}$. Consequently, eventually,

$$\ell_A(r_n) = -(y^*)^\top r + n\lambda_b + \frac{M_{n \bmod \tau_b}}{L}.$$

Proof. Choose finitely many $t_1, \dots, t_m \in \mathcal{S}_b$ with $\gcd \tau_b$. The additive semigroup they generate contains every sufficiently large multiple of τ_b . For each residue class s , choose n_s attaining the minimum M_s . If $n \equiv s \pmod{\tau_b}$ is sufficiently large, then $n - n_s$ is a nonnegative combination of the t_i . Appending the associated zero-defect completion shifts to an optimum at n_s produces a feasible completion at n with the same defect M_s . Minimality of M_s gives equality. $\square$

Since $t\lambda_b$ is integral for every $t \in \mathcal{S}_b$, Bezout's identity gives $\tau_b\lambda_b \in \mathbb{Z}$. If

$$\lambda_b = a/q$$

is reduced, then $q \mid \tau_b$.

8.2 The exact minimal eventual affine period

View

$$M = (M_0, \dots, M_{\tau_b-1})$$

as a cyclic word, and define its fundamental circular period

$$d_M = \min\{p \geq 1 : M_{s+p} = M_s \ \forall s \pmod{\tau_b}\}.$$

Then $d_M \mid \tau_b$.

Theorem 8.2 (Exact minimal eventual distance period). *If $\lambda_b = a/q$ in lowest terms, the smallest positive integer p for which there is an integer D_p satisfying*

$$\ell_A(r_{n+p}) = \ell_A(r_n) + D_p$$

for all sufficiently large n is

$$p_{\min}(r, b) = \text{lcm}(q, d_M).$$

Moreover $p_{\min}(r, b) \mid \tau_b$ and $D_p = p\lambda_b$.

Proof. If p is an eventual affine period, division by p and the asymptotic formula in theorem 8.1 force $D_p = p\lambda_b$, so $q \mid p$. The same formula then shows that the periodic correction must satisfy $M_{s+p} = M_s$ for every s , hence $d_M \mid p$.

Conversely, for $p_0 = \text{lcm}(q, d_M)$, the quantity $p_0\lambda_b$ is integral and the defect profile is invariant under shift by p_0 . The formula of theorem 8.1 therefore gives

$$\ell_A(r_{n+p_0}) = \ell_A(r_n) + p_0\lambda_b$$

for all sufficiently large n . Since both q and d_M divide τ_b , so does p_0 . $\square$

8.3 Eventual local geodesic periods

For $J \subseteq [d]$ put

$$c_J = A\mathbf{1}_J.$$

Apply theorem 8.1 to each shifted ray $r_n + c_J$. The same dual certificate, zero-defect semigroup, and canonical period τ_b apply; only the finite stabilized defect table changes. By theorem 3.2,

$$J \in \mathcal{H}_A(r_n) \iff \ell_A(r_n + c_J) = \ell_A(r_n) - |J|.$$

Thus the complete finite complex $\mathcal{H}_A(r_n)$ is eventually a function only of $n \bmod \tau_b$.

Let Θ_s denote this stabilized complex on residue class s , and set

$$d_{\mathcal{H}} = \min\{p \geq 1 : \Theta_{s+p} = \Theta_s \ \forall s \pmod{\tau_b}\}.$$

Corollary 8.3 (Exact minimal local-geometry period). *The smallest eventual period of the local action complex is*

$$\boxed{p_{\min}^{\mathcal{H}}(r, b) = d_{\mathcal{H}},}$$

and $d_{\mathcal{H}} \mid \tau_b$. The first-action set is eventually periodic by the same argument.

The point is not generic semilinear periodicity, which is classical, but that one minimum-completion defect mechanism simultaneously governs numerical distance and the local shortest-geodesic observables proved above.

Remark 8.4 (Multidirectional scope). *The same zero-defect construction can be applied generator-wise inside a finitely generated cone whose directions share a common dual-optimal face, leading to a finite residue-torus description. A full multidirectional statement is not needed for any theorem in this paper and is therefore not formulated here; this also keeps the manuscript separate from the established chamberwise theory of vector partitions and parametric integer programming [11].*

9 Related work and scope of contribution

Affine and half-factorial factorization theory. The augmented columns $C_i = (A_i, 1)$ form a homogeneous, hence half-factorial, affine semigroup. Factorization lengths, catenary-type invariants, and effective computations in half-factorial affine semigroups are established subjects [1]. Our optimal fiber is a particular equal-length factorization fiber; this algebraic fact is background.

Graver and Markov moves. Primitive conformal lattice moves originate in integer programming and Graver-basis theory [2], while finite move sets connecting fibers are fundamental in algebraic statistics and Markov-basis theory [3]. We use the equal-length kernel to describe collisions, but do not claim that finite fiber moves are new.

Squarefree divisor complexes. The complex in theorem 5.1 is a standard squarefree divisor complex, a construction used to encode multigraded Betti data of semigroup rings and studied explicitly for numerical semigroups [4]. Our contribution is the geodesic interpretation and the rank-sensitive local/global comparison, not the simplicial-complex definition.

Factorization-derived graphs and cubical complexes. Graphs and complexes assembled directly from factorization fibers are also established objects. In particular, Moschetti and O'Neill study factorization support and trade graphs and exhibit a cubical complex arising from factorization data [5]. Thus neither attaching graphs to factorization fibers nor obtaining cubical topology from such data is claimed as new here; the distinction is the restriction to prefixes of minimum completions followed by the equal-length residual quotient.

Cubical state and rewriting complexes. State complexes encode commuting independent moves by cubes in reconfigurable systems [8]. Cubical complexes also arise naturally from rewriting and diagram-group constructions; Squier's work and the later diagram-group framework are central examples [9, 10]. Consequently, filling commuting moves by cubes is not itself a novelty claim. The object studied here is narrower: the cubical subspace generated only by prefixes of minimum completions, followed by the equal-length residual quotient.

Vector partitions and asymptotics. Piecewise quasipolynomial behavior of vector partition functions is classical [11]. Our asymptotic statements should therefore be read as exact project-specific refinements: the zero-defect semigroup supplies a canonical eventual period, the stabilized defect table identifies the minimal observable period, and the same table controls geodesic action data.

Shortest paths in counter systems. Shortest paths in one-counter systems have a broad algorithmic theory [12]. The present residual graphs are much more structured. The emphasis here is not a generic shortest-path bound but an exact description of all shortest-prefix schedules and their topology after residual identification.

10 Discussion

The binary rectangle survives in higher dimension only after a conceptual change. With two letters, the minimum Parikh demand is unique and order is the only source of multiplicity. For a matrix constraint, the minimum demand itself becomes a fiber. The finite object

$$\mathcal{C}_A(r) = \downarrow O_A(r)$$

retains every partial schedule compatible with at least one optimum, while the lattice

$$K_A = \ker_{\mathbb{Z}}[A; \mathbf{1}^\top]$$

records exactly the trades that preserve both residual and length. Their quotient is therefore the natural higher-dimensional replacement for the binary rectangle.

The rank-one theorem shows that even the first nontrivial equal-length trade direction changes global topology qualitatively. A nonunique fiber creates a circle in the filled shortest interval, although local action compatibility remains a union of contractible interval nerves. At rank two, local topology can itself develop holes. This separates two notions that are easy to conflate: local coexistence of shortest actions and global identification of shortest schedules.

The current results do not classify higher-rank global homotopy types. Nor do they claim a new general theory of affine-semigroup topology, Graver connectivity, cubical state spaces, or parametric integer programming. The principal open direction is to understand which homotopy

types can occur in $\mathfrak{I}_A(r)$ when $\text{rank } K_A \geq 2$, and whether useful chamberwise classifications of the full global quotient accompany the finite local periodicity established above.

11 Conclusion

For completion-total matrix counting languages, minimum completion has a finite geometric package. The optimal Parikh fiber determines a down-set of exactly the shortest prefixes; equal-length kernel translations identify precisely those prefixes that represent the same residual at the same geodesic depth; and the resulting cubical quotient is the filled shortest residual interval.

This perspective yields an exact literal-box criterion and a complete rank-one global topology: unique optimum gives a contractible box, while every nonunique rank-one optimum gives a space homotopy equivalent to S^1 . The local compatibility complex, identified with a standard squarefree divisor complex, remains componentwise contractible in rank at most one, whereas rank two already permits a local first-homology hole. In scalar systems this produces the sharp $d = 2, 3, 4+$ shortest-geodesic hierarchy. Along residual rays, the same optimization structure stabilizes into a finite defect profile that determines exact eventual distance and local-geometry periods.

The broad surrounding theories are classical; the intended contribution is the exact coupling of minimum-completion fibers, shortest-prefix schedules, residual quotients, and equal-length lattice topology.

A A compact proof dependency map

The main structural chain is

$$\boxed{\text{completion totality}} \implies \boxed{O_A(r)} \implies \boxed{\mathcal{C}_A(r) = \downarrow O_A(r)} \implies \boxed{\mathfrak{I}_A(r) = |\mathcal{C}_A(r)|/K_A}.$$

The rank-one branch is

$$\begin{aligned} \boxed{K_A = \mathbb{Z}g} &\implies \boxed{\text{arithmetic optimal fiber}} \implies \boxed{\text{contractible overlap complexes}} \\ &\implies \boxed{\text{contractible periodic saturation}} \implies \boxed{\mathfrak{I}_A(r) \simeq S^1 \text{ if nonunique}}. \end{aligned}$$

The asymptotic branch is

$$\begin{aligned} \boxed{\text{directional LP}} &\implies \boxed{\mathcal{S}_b, \tau_b} \implies \boxed{\text{stabilized defect profile } M} \\ &\implies \boxed{\text{exact minimal eventual periods}}. \end{aligned}$$

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Data and code availability. The paper is theoretical and uses no empirical dataset. Algebraic examples and formulas are explicit in the manuscript. Any accompanying consistency scripts will be distributed with the submission package if required by the selected venue.

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